

SQL AGGREGATES AND OPERATORS

The Student Table

Table Students

Student ID	Name	Age	Department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

①

Select department
from Student table;

List all distinct departments
in the students table

Name	Department
Alice	IT
Bob	HR
Charlie	IT
Diana	Finance
Eve	HR

② Get the average age of students per department

Select department
from average_age;
Student table

Department	Average
IT	20 ; 21
HR	22
Finance	23

③ Show departments with more than 1 student

Select department
Student count

Department	Student - Count
HR	2
IT	2

④ Get all the students whose age is between 21 and 23

Select X
from Student - table

Student ID	Name	Age	Department
3	Charlie	21	IT
2	Bob	22	HR
5	Eve	22	HR
4	Diana	23	Finance

2. The Courses Table

Course ID	Course Name	Department	Credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

(6)
 Select department
 total credits Sum. As Total Crls
 from Courses table
 Group By Department

Department	Total Credits
IT	11

⑦ List all Courses that not have 4 Credits

Select *
From Courses table;

Courses

ID	Course name	Department	Credits
104	Finance Excel	Finance	2
105	Statistics	HR	3

⑧ Show the top 3 Courses by Credits in descending order

Select Course ID
Course name
Credits
From Table Courses
order by Credits Desc;

ID	Course name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3
104	Excel	2

Q. 3. The Enrollments Table

Table Enrollments

enrollment ID	Student ID	Course ID	grade
1	1	101	95
2	2	102	78
3	3	103	90
4	4	104	78
5	5	105	92

Q. Get the maximum, minimum, and the average grade across all enrollments

Select Max (grade) As max-grade
From Enrollment table

Select min (grade) As min-grade
From Enrollment table

Select Avg -grade As avg-grade
From Enrollment table

Enrollment ID	Grade	
3	99	Max
2	78	Min
1	95	Avg

⑩ Count how many enrollments exist per course

Select
from Course ID
Enrollment table

Select
from Enrollment Count (as count)
Enrollment table

Course id	Enrollment id
101	1
102	2
103	3
104	4
105	5

4. SALARIES TABLE

Table Salaries

Employee id	Name	Department	Salary	Bonus
1	Tom	IT	6000	5000
2	Jerry	HR	5500	4000
3	Spike	Finance	10000	6000
4	Mike	IT	62000	5000
5	Butch	HR	54000	3500

SELECT department
 SUM (salary) AS - total salary
 FROM salary table
 GROUP BY department

SELECT department
 SUM (bonus) AS - total bonus
 FROM salary table
 GROUP BY Department

Department	Total Salary	Total Bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

(12) Show departments where average salary is above
 ₹ 55000

SELECT department
 SUM (salary) AS - avg salary
 FROM ~~total salary table~~ employees
 GROUP BY department
 HAVING AVG (salary) > 55000;

SELECT department
 SUM (salary) AS avg salary
 FROM salary table
 GROUP BY Department

Department	Avg Salary
IT	60000
IT	62000
Finance	70000

⑩ List all the employees whose salary plus bonus is greater than 60000

SELECT Employee Id,
name,
Salary,
bonus,
FROM Employees
WHERE (Salary + bonus) > 60000;

Employee Id	Name	Salary	Bonus	Total Compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tulce	62000	5500	67500

5. The Projects Table

(14) Show total and average budget per department

Project Id	Project name	Department	budget
1	AI App	IT	12000
2	Payroll System	Finance	8000
3	Dashboard	IT	150000
4	Website	Marketing	60000
5	HR Portal	HR	50000

Select

department

Sum(budget) AS total-budget

AVG(budget) AS avg-budget

From

projects

Group By

department

Having

AVS(budget) > 70000;

Department	Total Budget	Average Budget
IT	120000	Budget
Finance	80000	
IT	150000	

(15) List all projects with budgets between 50,000 and 120,000 excluding the marketing department

Search

project name,
department,
budget

from
where

project's
budget

Between 50,000 & 120,000

AND

department < > marketing

Project name	Department	Budget
AI App	IT	120,000
Payroll System	Finance	80,000
HR Portal	HR	50,000